

GUARDIANS OF TOMORROW

K.L.N. College of Engineering | BE CSE (IoT)

An Explainable, Threshold-Optimized XGBoost Framework for Parkinson's Disease Detection from Sustained-Vowel Vocal Biomarkers

Problem Statement ID: TRN-05	Theme: Health Care	Team ID: Tn001
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Team Guardians of Tomorrow, K.L.N. College of Engineering, BE CSE (IoT) Final Year

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Abstract

Parkinson's disease (PD) lacks an early, low-cost, non-invasive screening method. We present an end-to-end explainable ML pipeline that detects PD from sustained-vowel voice recordings using correlation filtering, L1 regularization, RFE, SHAP-based feature selection, SMOTE class balancing and Standard Scaler normalization. An XGBoost classifier using eight SHAP-selected biomarkers correctly identified 28 of 29 PD cases on held-out data (92.31% accuracy, 93.33% precision, 96.55% recall, 94.92% F1, 97.24% ROC-AUC), improving on the untuned baseline (82.05% accuracy, 88.62% ROC-AUC). Lowering the decision threshold from 0.50 to 0.10 further boosted sensitivity for screening. The static model is complementary to digital-twin concepts, which suit longitudinal monitoring but are not yet validated for PD.

Problem Statement

PD is usually diagnosed only after motor symptoms appear, and access to specialists is limited in many regions, so patients are diagnosed late. Voice changes accompany PD and can be captured with an ordinary microphone, but turning them into a dependable screening aid is hard:

- Redundant, high-dimensional acoustic features cause overfitting on small datasets.
- Class imbalance biases models towards the majority class.
- Black-box models are hard for clinicians to trust, and the default 0.50 threshold can miss PD cases.

Problem: To build and validate an explainable ML pipeline that screens for Parkinson's disease from a single sustained-vowel recording using a few interpretable vocal biomarkers, with a clinically calibrated threshold, so that it is high-sensitivity, transparent to clinicians, and suitable for early, low-cost, non-invasive screening.

SDG Mapping

Goal	Target	Contribution
SDG 3 Good Health and Well-being	3.4 Reduce premature mortality from NCDs; promote well-being. 3.8 Access to quality essential health-care.	Primary. Early, non-invasive, low-cost PD screening from voice; fewer missed cases and explainable output for clinicians.
SDG 9 Industry, Innovation and Infrastructure	9.5 Enhance scientific research and technological capabilities.	Supporting. Applies explainable AI and IoT-ready audio sensing to digital health innovation.